

Strike and Release: Proximity-Guided Transient-Contact Control for Robotic Percussion

Hongyi Yang and Chenxi Xiao[†]

Abstract—Robotic percussion requires accurately timed impacts followed by rapid mallet release to produce clear and repeatable tones. Conventional contact-feedback strategies can initiate retraction only after impact, and delays in sensing, communication, and control may prolong mallet–surface contact and suppress instrument vibration. Fixed joint-position triggering avoids post-contact sensing but is sensitive to calibration errors, trajectory tracking, and variations in the relative instrument geometry. To bridge the gap, we propose a proximity-guided transient-contact framework that initiates retraction relative to the observed instrument surface. Specifically, a wrist-mounted time-of-flight sensor monitors the final approach and generates a pre-contact event that reverses the commanded distal-joint velocity before sustained contact develops. Joint-space impedance control accommodates residual impact uncertainty, while minimum-jerk interpolation provides smooth transitions between successive ready configurations. The framework is evaluated on a Franka Research 3 robot performing five multi-note percussion sequences over 1,580 commanded strikes. Compared with fixed joint-position triggering, the proposed method improves the strike success rate from 74.18% to 97.34% and the valid strike success rate from 76.30% to 98.72%. Representative high-speed recordings and acoustic responses further demonstrate more rapid mallet separation and more sustained instrument vibration. These results show that pre-contact proximity triggering provides a simple and effective execution primitive for repeated transient-contact percussion.

I. INTRODUCTION

Anticipation is a central feature of skilled sensorimotor behavior. Rather than waiting for physical contact to occur, humans and animals often begin preparing the appropriate post-contact response in advance. Such prospective control is particularly valuable in transient-contact tasks, where the interaction duration is short and purely reactive feedback may arrive too late to shape the outcome. Robotic percussion is one such task. Percussion presents several demanding challenges for dynamic robotic interaction. To generate a clear sound, a successful strike requires precise contact timing and location, sufficient pre-impact velocity, and rapid release [1], [2]. In contrast, excessive stiffness can increase impact loads and inhibit mallet rebound, while delayed retraction can prolong mallet–surface contact and suppress the instrument’s free vibration, thereby degrading sound quality [1], [3].

To address these challenges, recent robotic musical systems have commonly employed custom percussion mechanisms or humanoid platforms, enabling improvised and coordinated performances [4]–[7]. Although these systems

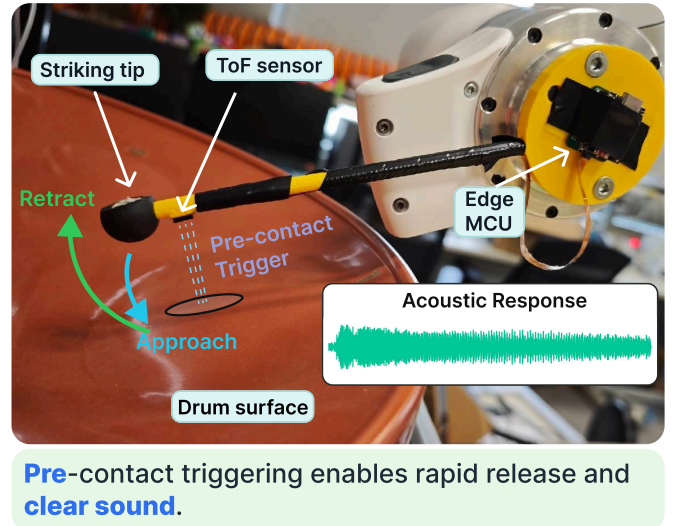


Fig. 1. Predictive transient-contact percussion using wrist-mounted proximity sensing. During the final approach, the ToF sensor monitors the drum surface and triggers retraction before contact, enabling rapid mallet release and clear sound production.

have addressed musical interaction, rhythmic coordination, and contact-rich stick manipulation across dedicated and general robotic platforms, the acoustically critical timing of mallet release during individual strikes remains insufficiently studied. Beyond the aforementioned requirements for timing, location, and velocity, our experiments reveal that variability in release timing is a critical determinant of sound quality during repeated strikes. Conventional contact-triggered strategies rely on joint-torque, force, or tactile feedback [8], [9], but evidence of contact in these signals becomes available only after impact begins. Sensing, filtering, communication, and control delays may therefore cause the mallet to remain pressed against the surface. By contrast, a preprogrammed trigger based on a fixed joint position avoids reliance on post-contact sensing, but its timing is sensitive to calibration errors, trajectory-tracking errors, and changes in the instrument pose. Moreover, such an approach requires manual adjustment: a conservative threshold may trigger retraction too early, whereas a deeper threshold may delay release and produce a damped sound.

To bridge this gap, we investigate percussion on a general-purpose robotic platform and formulate contact release as a *predictive release* problem [10]. Instead of relying on fixed action sequences or post-contact feedback, we use active perception to adapt each strike to the observed instrument surface. As illustrated in Fig. 1, the striking motion is gener-

[†] denotes the corresponding author.

Hongyi Yang and Chenxi Xiao are with the School of Information Science and Technology, ShanghaiTech University, Shanghai, China. {yanghy22022, xiaochx}@shanghaitech.edu.cn

Project webpage: <https://xuhaoqin02.github.io/lerobot-tactile/>

ated primarily through the robot’s distal joint, while a wrist-mounted time-of-flight sensor monitors the final approach. When the measured distance reaches a specified threshold, a pre-contact event immediately switches the commanded motion from approach to retraction. Although the command is reversed before physical contact, the remaining incoming momentum allows the mallet to complete the impact while promoting rapid separation from the surface. To accommodate uncertainty in the strike distance, impact, and relative instrument geometry, the striking motion is executed under joint-space impedance control, which absorbs residual collision energy and avoids excessively rigid impacts. Minimum-jerk trajectories are further used to transition smoothly between strike locations, reducing abrupt state changes and residual vibrations between successive strikes.

Experiments across five multi-note sequences and 1,580 commanded strikes demonstrate that predictive proximity triggering improves the aggregate strike success rate from 74.18% to 97.34%, an absolute gain of 23.16 percentage points over fixed joint-position triggering. The valid strike success rate similarly increases from 76.30% to 98.72%, while minimum-jerk transitions reduce the median terminal joint-velocity RMS from 0.1553 rad/s to 0.0049 rad/s over 122 paired transitions. These results support the use of the proposed framework in robotic percussion and other rapid transient-contact tasks.

The contributions of this work are as follows:

- We investigate repeated percussion on a general-purpose robotic platform and identify timely mallet release as essential to reliable strike execution and sound production.
- We develop a wrist-mounted proximity-sensing module and a predictive, proximity-triggered retraction strategy that enables timely mallet release.
- We validate the proposed framework on a torque-controlled manipulator over more than 1,580 commanded strikes and analyze strike reliability and residual motion between successive strikes.

II. RELATED WORK

A. Robotic Musical Performance

Robotic musical performance investigates how robots perceive, generate, and physically execute musical structures. It provides a challenging testbed for rhythmic synchronization, repeated high-speed motion, and coordinated physical interaction with instruments. Early systems such as Haile and Shimon focused on machine listening, improvisation, and human–robot collaboration using dedicated musical robots [1], [4], [5], [11]. More recent studies have increasingly adopted learning-based methods [12]. Karbasi et al. used intrinsically motivated reinforcement learning to train underactuated robotic drummers, showing that the resulting rhythmic patterns were shaped by both the learning process and the robots’ physical dynamics [13]. Robot Drummer represented percussion scores as timed contact sequences and learned long-horizon humanoid drumming policies through

reinforcement learning [6]. DexDrummer further combined trajectory planning and residual reinforcement learning to address bimanual coordination, in-hand drumstick control, and contact-rich long-horizon performances [7].

These systems represent substantial progress in high-level rhythm execution, whole-body coordination, and dexterous musical manipulation. However, they primarily treat a strike as one element of a planned or learned action sequence, rather than explicitly studying the low-level timing required to terminate contact immediately after impact. Consequently, they do not isolate the delay introduced when retraction is triggered only after joint or contact feedback indicates that impact has already occurred. In contrast, our work does not aim to outperform these methods in musical intelligence or dexterity. Instead, it introduces a lightweight contact-execution primitive that bridges high-level strike commands and reliable transient-contact realization: a wrist-mounted proximity sensor predicts imminent contact, while compliant torque control absorbs residual impact uncertainty and enables rapid retraction.

B. Impact-Aware Manipulation

Impact-aware manipulation studies robotic interactions in which rapid contact transitions produce impulsive forces, velocity discontinuities, and potentially unstable feedback responses. Its objective is to regulate robot motion before, during, and after impact while maintaining task performance and mechanical safety [3], [14]. Khurana and Billard jointly introduced the hitting flux metric and a constrained control framework that enables robots to execute repeatable strikes with desired contact direction, speed, and inertia [15]. Reference-spreading control connected pre-impact and post-impact vector fields through an impact model, reducing feedback peaks during contact transitions [16], [17]. More recently, IMA-Catcher combined pre-impact motion optimization with learned post-impact energy dissipation for nonprehensile catching of moving objects [18]. These approaches provide more general and theoretically sophisticated impact-handling mechanisms than the method proposed in this paper.

Nevertheless, most existing methods target momentum regulation, object catching, or stable post-impact interaction, and often require impact models, object-state estimation, or online trajectory optimization. Such formulations are not directly designed for repeated percussion, where the key objective is not to establish or stabilize contact, but to prevent the mallet from remaining on the instrument surface after a short impact. Purely reactive impedance or torque feedback also cannot eliminate the causal delay of detecting contact after it has occurred. Our method bridges this gap by using a pre-contact distance event to initiate retraction before sustained loading develops. The resulting framework is less general than model-based impact optimization, but is computationally lightweight, interpretable, and directly applicable to repeated transient-contact tasks executed by a general-purpose torque-controlled manipulator.

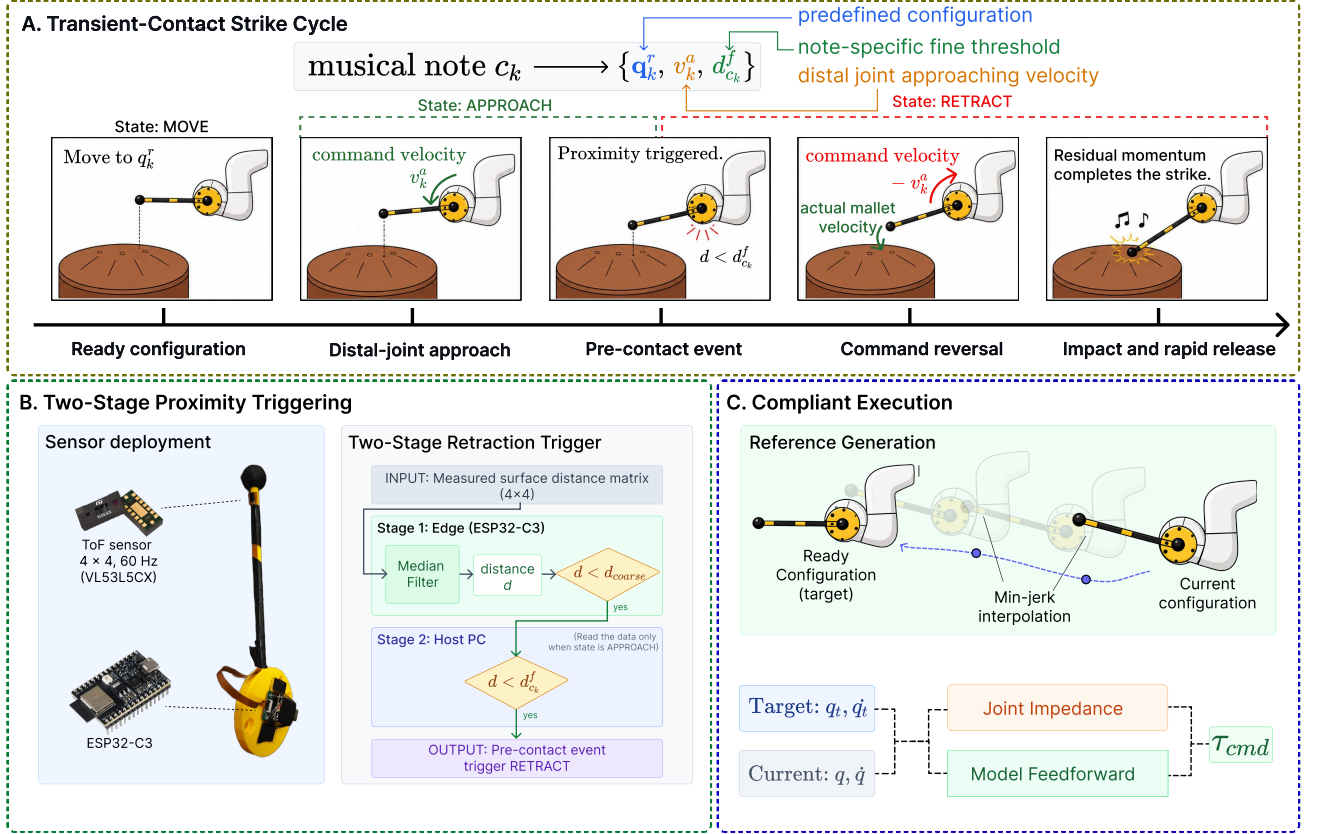


Fig. 2. System overview of the proposed transient-contact percussion framework. (A) Each commanded strike transitions through MOVE, APPROACH, and RETRACT, with a pre-contact event reversing the commanded distal-joint velocity before impact. (B) Wrist-mounted ToF sensing and edge-side filtering generate a surface-relative proximity event. (C) Minimum-jerk interpolation provides smooth transitions between ready configurations, while a 1-kHz joint-space impedance controller executes the latest motion reference.

III. METHOD

A. Problem Formulation and System Overview

We consider a percussion sequence containing N commanded strikes,

$$\mathcal{S} = \{q_k^r, v_k^a, d_{c_k}^f\}_{k=1}^N, \quad (1)$$

where $q_k^r \in \mathbb{R}^7$ is the ready configuration associated with the target note, v_k^a is the approach velocity assigned to the distal joint, and $c_k \in \{1, \dots, 8\}$ denotes the target-note index. The corresponding host-side note-specific fine threshold is denoted by $d_{c_k}^f$. The remaining joints are regulated around q_k^r during striking. Rather than maintaining contact or regulating a sustained contact force, the objective is to produce a short impact and initiate retraction before the mallet remains pressed against the instrument surface.

As illustrated in Fig. 2, the proposed framework separates offline strike planning, low-rate proximity sensing, and high-rate robot control. The target sequence is first converted into ready configurations and strike velocities. During execution, the host computer receives proximity events and updates a finite-state motion command, while the robot controller executes the latest reference through a 1-kHz torque-control loop.

Each strike is organized into three controller states—MOVE, APPROACH, and RETRACT—and five conceptual

moments, as illustrated in Fig. 2. First, during MOVE, the robot transitions from the previous configuration to q_k^r , while proximity triggering remains inactive. Second, the controller enters APPROACH, and the distal joint drives the mallet toward the instrument while the proximity sensor monitors the surface. Third, a valid pre-contact event switches the controller from APPROACH to RETRACT. The commanded distal-joint velocity is then reversed, although the actual mallet velocity may remain directed toward the surface briefly because of inertia and controller compliance. This residual incoming momentum completes the impact, while the early command reversal promotes rapid separation immediately afterward. The controller remains in RETRACT until joint 7 reaches the predefined withdrawal angle, after which it enters MOVE for the next target.

B. Two-Stage Proximity Triggering

A VL53L5CX time-of-flight sensor is mounted near the robot wrist and oriented toward the instrument surface. It provides a 4×4 distance array at 60 Hz [19]. Fixed joint-position triggering cannot directly account for variations in the relative mallet–surface geometry, motivating the use of target-relative proximity triggering. Since peripheral zones may observe the mallet, drum boundary, or background, only a predefined set of valid zones is retained. Invalid returns are

discarded, and the median distance of the valid measurements is computed on an ESP32-C3 to suppress isolated sensing errors.

Proximity detection is divided into an edge-side coarse gate and a host-side fine trigger. The ESP32-C3 continuously filters the distance array and computes the median valid-zone distance d . When $d < d_{\text{coarse}}$, the device transmits the filtered distance as a compact two-byte integer rather than continuously streaming the raw 4×4 array.

After receiving the packet, the host selects the note-specific fine threshold $d_{c_k}^f$ associated with the current target and accepts the event only when the measured distance satisfies $d < d_{c_k}^f$ and the robot is in the APPROACH state. A valid event immediately changes the state to RETRACT, updates the motion command, and returns a one-byte acknowledgment to the embedded device. Further events are ignored until the next strike, preventing repeated switching during the same contact cycle.

The sensing pipeline is decoupled from the high-rate robot controller: the sensor produces discrete events at 60 Hz, whereas the latest command is executed at 1 kHz. The trigger represents imminent contact rather than the physical impact instant. Therefore, retraction can begin before force or torque feedback becomes available, compensating for sensing, communication, and command-update delays while preserving sufficient momentum to complete the strike.

C. Strike Motion and Compliant Execution

Minimum-jerk interpolation is used only for transitions between successive ready configurations [20]. Without smooth interpolation, abrupt configuration changes may introduce large acceleration and jerk, leading to inconsistent initial conditions for the next strike. Zero terminal velocity and acceleration are imposed so that each approach begins from a smooth and repeatable state. The approach and retraction phases instead use task-defined distal-joint velocities to preserve the required impact dynamics.

The motion is executed by a 1-kHz joint-space impedance controller with model-based feedforward compensation [21]. During each strike, the proximal joints remain regulated near the selected ready configuration, while the distal joint drives the mallet toward the instrument. Once a valid proximity event is received, its commanded velocity is immediately reversed to initiate retraction. Impedance control accommodates small tracking and geometric errors during the residual impact, whereas the proximity event determines the release timing.

IV. EXPERIMENTS

A. Experimental Setup and Protocol

Experiments were conducted on a Franka Research 3 manipulator equipped with a customized rubber-tip mallet and a wrist-mounted VL53L5CX multi-zone ToF sensor, as shown in Fig. 3. The robot struck a 40-cm steel tongue drum under joint-space torque control executed at 1 kHz through FCI and Polymetis. The sensor operated in a 4×4 mode at 60 Hz with a 5-ms integration time. The edge-side coarse

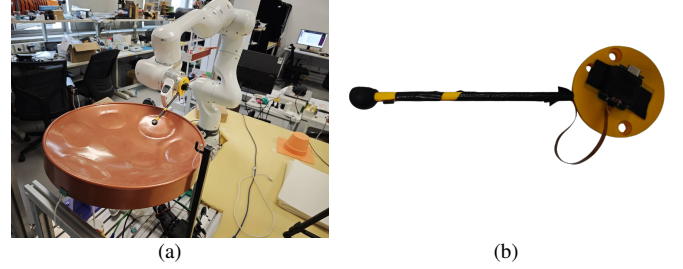


Fig. 3. Experimental platform. (a) Franka Research 3 percussion setup with a steel tongue drum and customized mallet. (b) Wrist-mounted VL53L5CX proximity sensor and mallet assembly.

TABLE I

NOTE-SPECIFIC TRIGGERING PARAMETERS. C1–C7 REPRESENT THE C-MAJOR SCALE DEGREES, WHILE D1 REPRESENTS THE D-MAJOR TONIC.

Target	Position threshold (rad)	Fine threshold (mm)
C1	-0.5715	15
C2	-0.5413	35
C3	-0.6408	17
C4	-0.6080	30
C5	-0.6214	30
C6	-0.6144	23
C7	-0.6406	20
D1	-0.6467	22

threshold was fixed at $d_{\text{coarse}} = 70$ mm, while the host-side fine threshold $d_{c_k}^f$ was calibrated separately for each target note, as illustrated in Table I.

The robot performed five children’s songs, each repeated five times under two retraction conditions. The *position trigger* reversed joint 7 at a pre-calibrated note-specific joint-angle threshold, whereas the proposed *proximity trigger* initiated retraction from the corresponding note-specific ToF distance event. Before evaluation, the joint-angle and proximity thresholds were calibrated separately for each of the eight target notes using preliminary trials. The thresholds were adjusted to achieve reliable surface contact while avoiding premature retraction and prolonged mallet–surface loading. The calibration trials were excluded from the reported results, and all thresholds were fixed before the evaluation runs. The same calibration objective and comparable tuning effort were used for both triggering methods. The note-specific triggering parameters are summarized in Table I. The note-specific ready configurations were manually calibrated before evaluation to align the mallet with the corresponding target regions. They were held fixed throughout all evaluation runs. Impedance gains, ready configurations, and approach and retraction velocities were identical between the two triggering conditions. Each condition contained 790 commanded strikes.

A strike was successful only when the mallet contacted the intended region and produced a clear, non-muffled sound. Strike outcomes were manually labeled from synchronized video and audio recordings by one annotator. Video recordings were used to verify contact with the intended target re-

gion, while audio recordings were used to determine whether the resulting tone was clear and non-muffled. We report the end-to-end strike success rate (SR) and a diagnostic valid strike success rate (VSR):

$$\text{SR} = \frac{N_{\text{success}}}{N_{\text{total}}} \times 100\%, \quad \text{VSR} = \frac{N_{\text{success}}}{N_{\text{total}} - N_{\text{network}}} \times 100\%. \quad (2)$$

VSR excludes strikes affected by logged Polymetis/FCI interruptions, while SR remains the primary metric. The benchmark evaluates strike execution rather than musical timing because the method does not prescribe inter-strike intervals.

B. Qualitative Analysis of Release Timing and Sound Quality

To qualitatively examine how mallet release timing affects the resulting sound, we conducted an additional study using only the position-triggered controller. For the same target note, the robot configuration, impedance gains, and approach and retraction velocities were held fixed, while the joint-position threshold for command reversal was varied. One threshold initiated retraction earlier along the approach trajectory, whereas the other delayed command reversal and resulted in a longer mallet-surface contact interval. Representative 420-fps high-speed recordings and synchronized acoustic responses are shown in Fig. 4.

The representative high-speed sequences suggest that earlier command reversal promotes outward mallet motion shortly after impact and enables more rapid separation from the drum surface. In contrast, later command reversal allows the mallet to remain in contact with the surface for a longer interval. The corresponding acoustic responses exhibit the same qualitative trend: the shorter-contact example shows more sustained oscillation, whereas the longer-contact example decays more rapidly. This observation supports the premise that timely mallet release is important for avoiding vibration suppression and producing a clear tone. Because the contact boundaries were manually identified from representative trials, the comparison is intended only as a qualitative illustration of the physical mechanism and not as a statistical estimate of contact duration or acoustic performance.

C. Position Trigger Versus Proximity Trigger

Table II reports the strike results for all five songs. The proposed trigger improves SR for every sequence, with absolute gains between 18.57 and 29.29 percentage points.

Aggregated over all strikes, proximity triggering increases SR from 74.18% to 97.34%, an absolute improvement of 23.16 percentage points. The corresponding VSR values increase from 76.30% to 98.72%, indicating that the improvement remains after logged network failures are excluded. Across the 25 runs, the mean per-run SR improves from $73.76 \pm 7.65\%$ to $97.34 \pm 2.85\%$. The lower variation is consistent with the hypothesis that direct surface-distance observation reduces sensitivity to geometric and tracking variations compared with a fixed joint-angle threshold. Dedicated perturbation experiments would be required to quantify the contribution of individual error sources.

D. Minimum-Jerk Transition Ablation

We further compared linear interpolation with the proposed minimum-jerk transition between strike locations. The proximity trigger, impedance gains, ready configurations, and all remaining control parameters were held constant. Joint velocities were recorded at 1 kHz during each MOVE phase, and transitions were paired by song and segment index. We excluded segments shorter than 100 samples, with total joint displacement below 0.03 rad, or with mismatched endpoints or durations, yielding 122 valid pairs.

Residual motion before the following strike was measured using the terminal velocity RMS over the final 100 ms:

$$V_s^{\text{term}} = \sqrt{\frac{1}{7|\mathcal{W}_s|} \sum_{i=1}^7 \sum_{n \in \mathcal{W}_s} \dot{q}_{i,s}^2[n]}, \quad (3)$$

where $\mathcal{W}_s$ denotes the terminal window of transition s . A smaller value indicates less residual joint motion before APPROACH.

Minimum-jerk planning reduces terminal residual velocity for all five sequences. Across the 122 paired transitions, the median segment-wise relative reduction is 96.41%. Separately, the distribution medians decrease from 0.1553 rad/s for linear interpolation to 0.0049 rad/s for minimum-jerk interpolation. These results indicate that minimum-jerk transitions provide a more stable initial condition before the controller enters APPROACH.

V. CONCLUSION, DISCUSSION, AND FUTURE WORK

This paper presented a proximity-guided transient-contact framework for repeated robotic percussion. The proposed system combines pre-contact surface sensing, compliant control, and trajectory planning to produce short impacts followed by rapid mallet separation. Experiments on five multi-note sequences showed that proximity-triggered retraction increased the aggregate strike success rate from 74.18% to 97.34% compared with fixed joint-position triggering. Minimum-jerk transitions further reduced residual motion before successive strikes. These results suggest that reliable robotic percussion depends not only on reaching the intended target, but also on anticipating imminent contact and issuing the retraction command early enough to promote rapid separation after impact. Compared with fixed joint-position thresholds, local distance sensing provides a more direct and reliable surface-relative trigger and better accommodates variations in the relative distance between the robot and the instrument surface.

The current system nevertheless has several limitations. First, it has been evaluated primarily on short, structured percussion sequences and does not yet address expressive intensity control or the accumulation of timing errors during long-horizon performances. Second, the proximity threshold and retraction distance still require manual adjustment. Future work will investigate learning these parameters from data and adapting them online to different instruments and striking conditions. We also plan to extend the framework

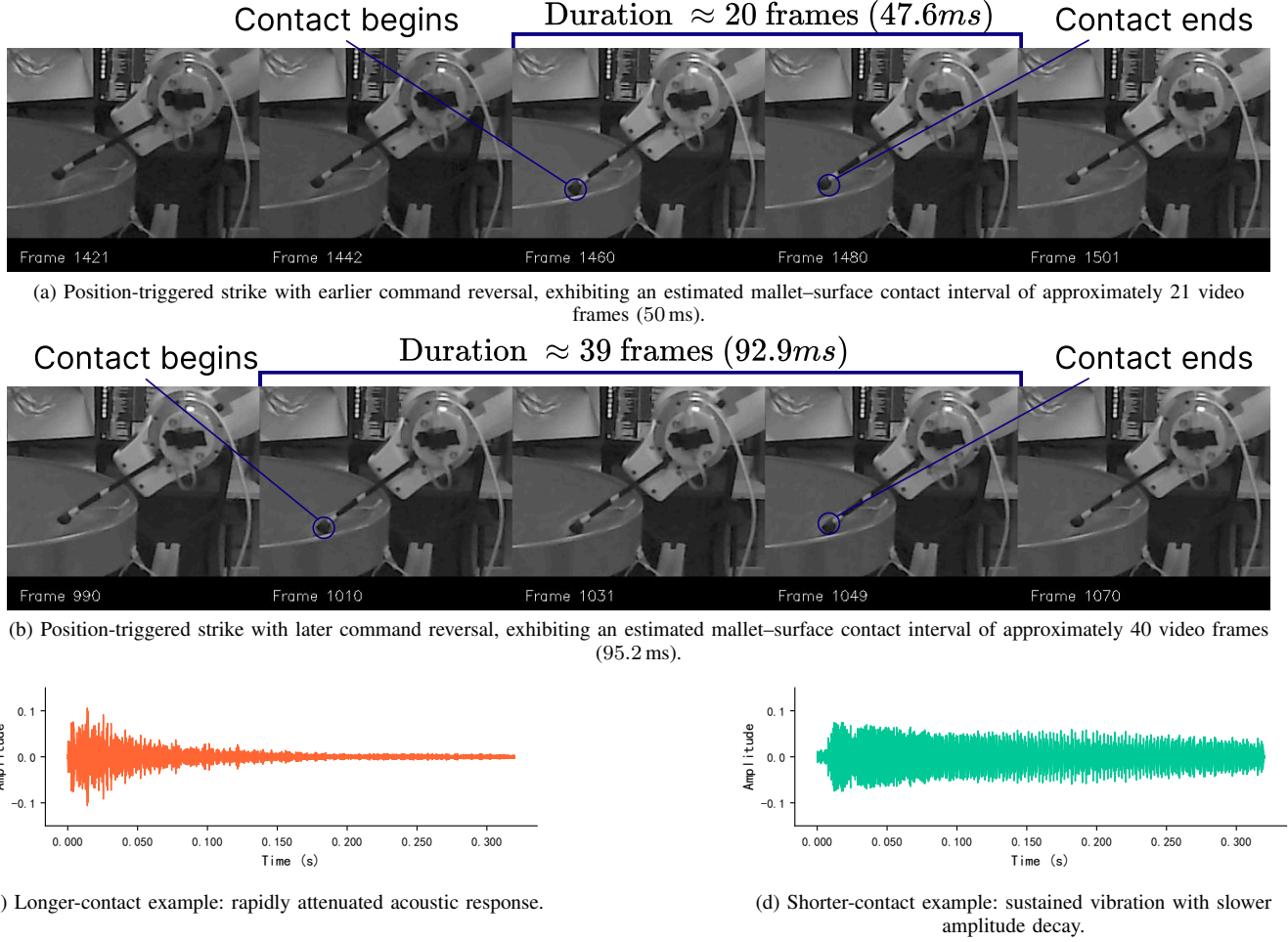


Fig. 4. Qualitative effect of position-triggered command-reversal timing on transient-contact behavior and acoustic response. Both examples use the same position-triggered controller, target note, and motion parameters; only the joint-position threshold for command reversal is varied. Earlier command reversal produces a shorter estimated mallet-surface contact interval and a more sustained oscillatory response, whereas later command reversal results in longer contact and more rapid amplitude decay. The contact boundaries are manually identified from representative trials and are used only to illustrate the physical effect of release timing, rather than to provide a population-level contact-duration measurement.

TABLE II
STRIKE SUCCESS UNDER POSITION- AND PROXIMITY-TRIGGERED RETRACTION. PARENTHESES SHOW SUCCESSFUL STRIKES OVER EVALUATED STRIKES.

Target sequence	Position trigger		Proximity trigger		Δ SR (p.p.)
	SR	VSR	SR	VSR	
Twinkle Twinkle	78.10% (164/210)	80.00% (164/205)	96.67% (203/210)	98.07% (203/207)	18.57
School Song	72.00% (90/125)	73.77% (90/122)	96.80% (121/125)	98.37% (121/123)	24.80
Looking for Friends	68.57% (96/140)	71.11% (96/135)	97.86% (137/140)	99.28% (137/138)	29.29
Kindergarten	75.65% (87/115)	79.09% (87/110)	97.39% (112/115)	98.25% (112/114)	21.74
Drop a Handkerchief	74.50% (149/200)	76.02% (149/196)	98.00% (196/200)	99.49% (196/197)	23.50
Aggregate	74.18% (586/790)	76.30% (586/768)	97.34% (769/790)	98.72% (769/779)	23.16

toward reproducing complete human drumming demonstrations by jointly exploiting multimodal visual, acoustic, proprioceptive, and contact signals.

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TABLE III

MEDIAN TERMINAL VELOCITY RMS FOR PAIRED TRANSITIONS.

REDUCTION DENOTES THE MEDIAN SEGMENT-WISE RELATIVE

REDUCTION AND IS NOT CALCULATED DIRECTLY FROM THE DISPLAYED
MEDIAN.

Sequence	Linear (rad/s)	Min.-jerk	Reduction (%)
Looking for Friends	0.1553	0.0057	96.58
Drop a Handkerchief	0.1538	0.0045	97.13
Kindergarten	0.1581	0.0055	96.26
School Song	0.1591	0.0093	94.80
Twinkle Twinkle	0.1597	0.0043	96.39
All transitions	0.1553	0.0049	96.41

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